

Investigating Galaxy Morphology using the HPP telescope

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Abstract

This report presents the results of investigating the morphology of three galaxies using the HPP telescope by fitting their measured intensity distribution with theoretical models. The galaxies studied were M81 (Bode Galaxy), M51 (Whirlpool Galaxy), and M49. GalFit was used to fit Sersic profiles to the galaxies to determine surface brightness, effective radius, and Sersic index. The results revealed the diverse morphologies of the galaxies, with M81 and M51 showing characteristics of spiral galaxies and M49 exhibiting an elliptical morphology. Here, the fit of the M51 galaxy agreed especially well with observations, while the fits of the other two galaxies appear to agree less well with expectations.

Contents

1	Introduction	2
2	Methodes	3
2.1	Dark Subtraction and Flat Fielding Correction	3
2.2	Cosmic ray rejection and background subtraction	3
2.3	Astrometric Calibration	4
2.4	Stacking and photometric calibration	4
2.5	GalFit	4
3	Results	5
3.1	Whirlpool galaxy	5
3.2	Bode Galaxy	6
3.3	M49	7
4	Discussion	8
4.1	Whirlpool Galaxy	8
4.2	Bode Galaxy	9
4.3	M49	9
5	Conclusion	9

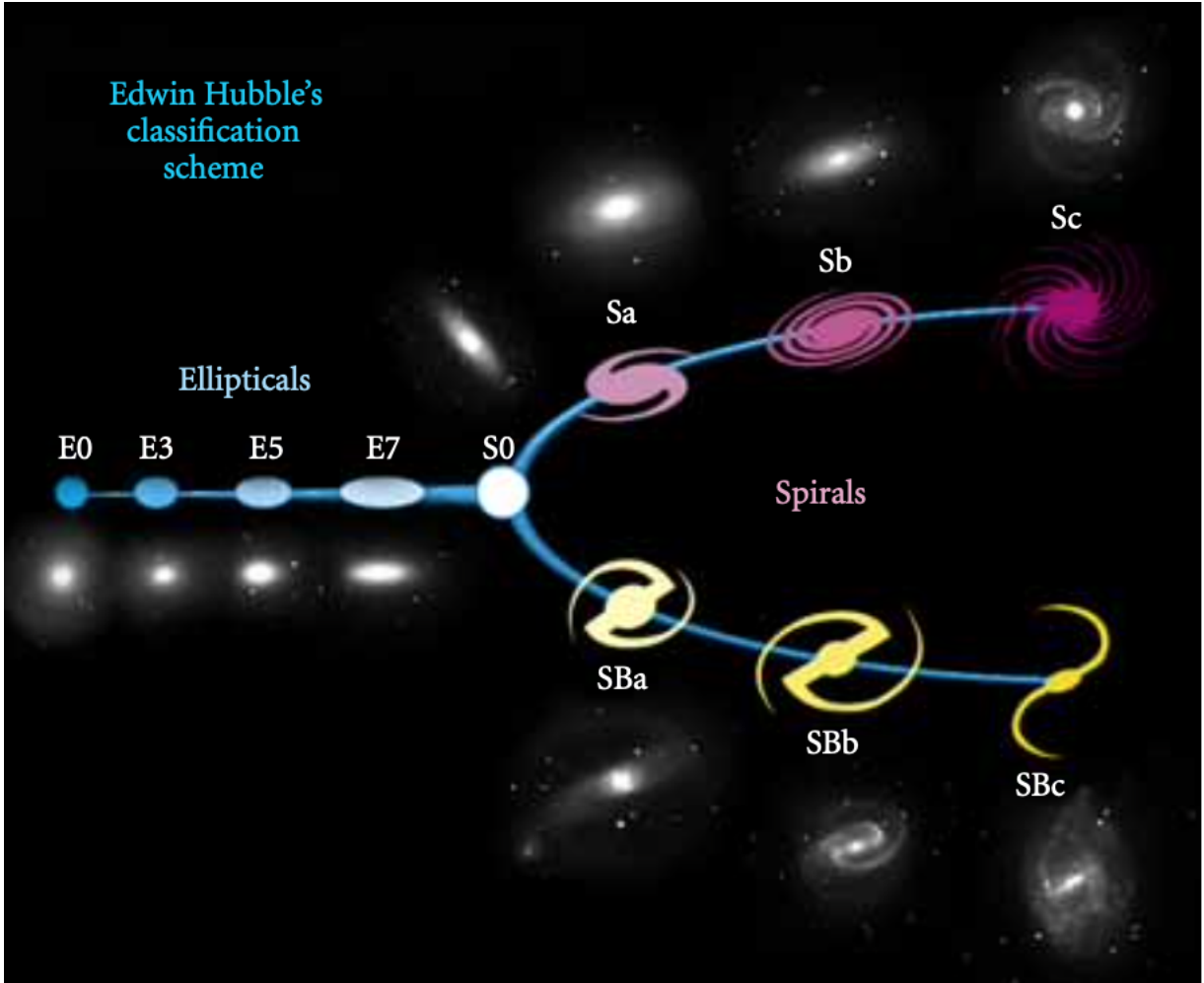


Figure 1: A depiction of the galaxy morphology classification system by Edwin Hubble[1].

1 Introduction

The study of galaxies and their morphologies provides us with insights into the processes that guide the evolution of the universe. Understanding galaxy morphology not only uncovers the properties of these objects but also contributes to our understanding of the larger universe. A depiction of the several types of galaxies can be seen in figure 1. Here, we see the the two general types of galaxies, elliptical and spiral which can be further divided into different types.

In this report, we present the results of an experiment aimed at investigating the morphology of three galaxies. To conduct our observations, we used the Planewave CDK20 telescope.

To analyze the collected data and classify the observed galaxies, we used the program GalFit. GalFit is a software package used in astrophysics that utilizes a Sersic profile

fitting technique. This approach enables us to model the surface brightness distribution of a galaxy and determine the specific morphological type of each observed galaxy.

The Sersic profile, is a mathematical function that describes the radial light distribution in galaxies. By fitting the Sersic profile to our observational data, we can determine the morphological parameters, such as the shape, size, and concentration of each galaxy. This information is important for classifying galaxies into elliptical, spiral, and irregular galaxies.

The rest of this report will explain our experimental methods, data acquisition and processing procedures. We will then introduce the process of fitting the Sersic profile using GalFit and present our findings regarding the morphological classification of the observed galaxies.

2 Methodes

Our experiment aimed to explore the morphologies of three different galaxies using the Planewave CDK20 telescope and GalFit. To do so we measured the intensities of the galaxies in the Johnson R-band. We chose the galaxies M81 (also called the Bode Galaxy), M51 (which is also known as the Whirlpool Galaxy) and M49. Here, the first two are spiral galaxies and the last one is a elliptical galaxy. We chose these to have a more diverse group of galaxies to try GalFit on. We observed each galaxy in the red filter of the telescope for a total integration time of 20 minutes. Before we then were able to use GalFit on these images we had to do some data reduction steps.

2.1 Dark Subtraction and Flat Fielding Correction

We began by creating a science image. The science frame is defined as

$$\text{science frame} = \frac{\text{raw science frame} - \text{master dark}}{\text{master flat} / < \text{master flat} >} \quad (1)$$

where the raw science frame is the exposure we took with the telescope of our target. The master dark is the median image of all the dark frames. A dark frame is an image taken with a closed camera shutter at the same exposure time as our science frame. With it we can reduce the dark current which is the noise caused by the camera electronic. The master flat is the median of all the flat images from which we subtract the dark for flats. A flat image is taken by pointing the telescope at a uniformly lit patch of sky during night fall. It corrects for uneven illumination and dust artifacts in the images taken by a camera or telescope.

2.2 Cosmic ray rejection and background subtraction

After creating the science image we then removed the impact of cosmic rays. For that we used the `detect_cosmics()` function from the Astroscrappy python package. The package

detects cosmic rays by using Pieter van Dokkums L.A.Cosmic algorithm [7]. We then move on to background subtraction which removes the contribution of unwanted signals or sources from our astronomical observations. For that we have to estimate the mean background for each science image and again used python to subtract it from the data.

2.3 Astrometric Calibration

We then move on to astrometric calibration, which is the process of finding a coordinate system for our data. For that we have to determine the positions of our galaxies within the images with respect to a known reference frame. To performed this calibration we then used the website astrometry.net.

2.4 Stacking and photometric calibration

Then we went on to stack the twenty science frames we created for each galaxy to create a science image. To do so we aligned the images using the astroalign python package. We then calculated the mean of all science images. Lastly we did the photometric calibration to calculate the photometric zero-point which we need for GalFit. With the photometric calibration we can convert the measured intensity into physical units, which in this case is magnitudes. For that we summed the flux of the galaxy and compare it with the magnitude of a reference star. In our experiment we used three different reference stars:

- HD 85-458 for the Bode galaxy
- TYC 3463-58-1 for M51
- Denebola (HR 4534) for M49.

We were then able to calculate the magnitude photometric zero-point, which is the offset between the literature magnitude and the measurements with the formula

$$m_{\text{literature}} = m_{zp} - 2.5 \log_{10} \left(\frac{N}{t} \right) \quad (2)$$

where N is the number of exposures and t is the exposure time.

2.5 GalFit

GalFit is a program which gives us the possibility to fit a Sersic model onto our galaxy to determine its morphology. Equation 3 shows the Sersic equation.

$$\Sigma(r) = \Sigma_e \exp \left[-\kappa \left(\left(\frac{r}{r_e} \right)^{1/n} - 1 \right) \right] \quad (3)$$

In this experiment we were especially interested in the parameters Σ_e called the pixel surface brightness, r_e which is the half-light radius and n which is called the concentration parameter or also the Sersic index. This Sersic index gives us the possibility to determine the morphology of our galaxies.

Objects with Sersic indices of less than 1 have profiles that are centrally concentrated, often with a central bulge. When the Sersic index is close to 1, the profile is exponential, which indicates a disk-structure with a relatively smooth decline in brightness as the radius increases. For Sersic indices around 4, the profile is a de Vaucouleurs profile, which typically describes elliptical galaxies. These profiles have a strong central concentration and faster decline in brightness towards the outside. Objects with Sersic indices greater than 4 generally show highly extended profiles like spiral arms or irregular structures. As Galaxies have different structural components, such as bulges, disks, bars, or spiral arms, one often needs to fit multiple sersic functions onto a galaxy.[2] [6] [5] [3]

3 Results

3.1 Whirlpool galaxy

With the reference star and equation 2 we calculated the magnitude photometric zeropoint which we determined to be $m_{zp} = 18.76 \pm 0.05$. Figure 2 shows the galaxy, the model which we fitted and the residual of the image after the model has been subtracted. The model shown here is a combination of four different Sersic functions.

Table 1: Sersic Model Parameters for the Whirlpool galaxy

Model	Integrated Magnitude	Effective Radius (pixels)	Sersic Index
1st Function	6.4285 ± 0.0251	25.6849 ± 0.2167	0.4117 ± 0.0107
2nd Function	4.8033 ± 0.4572	171.9334 ± 85.8887	4.5496 ± 0.4748
3rd Function	3.3790 ± 0.0111	255.5811 ± 1.5991	0.1574 ± 0.0035
4th Function	4.7806 ± 0.2738	120.7271 ± 34.7358	3.9602 ± 0.2584

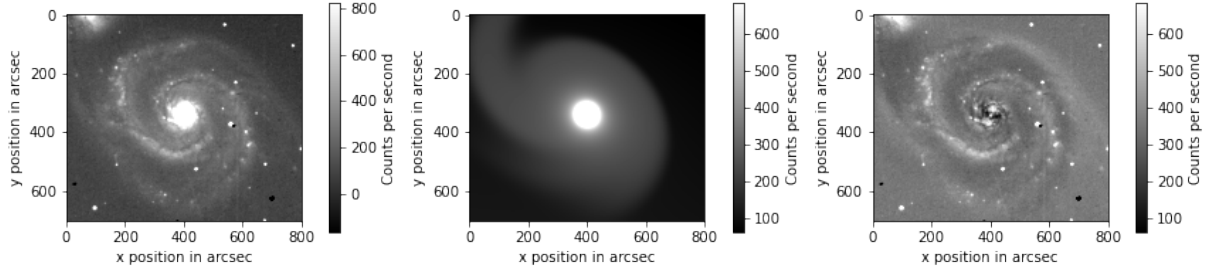


Figure 2: The Galfit result of the fitted Whirlpool galaxy. Here, one can see the galaxy, the fitted model and the residual.

3.2 Bode Galaxy

With the reference star and equation 2 we were then able to calculate the magnitude photometric zeropoint which we determined to be $m_{zp} = 21.01 \pm 0.005$.

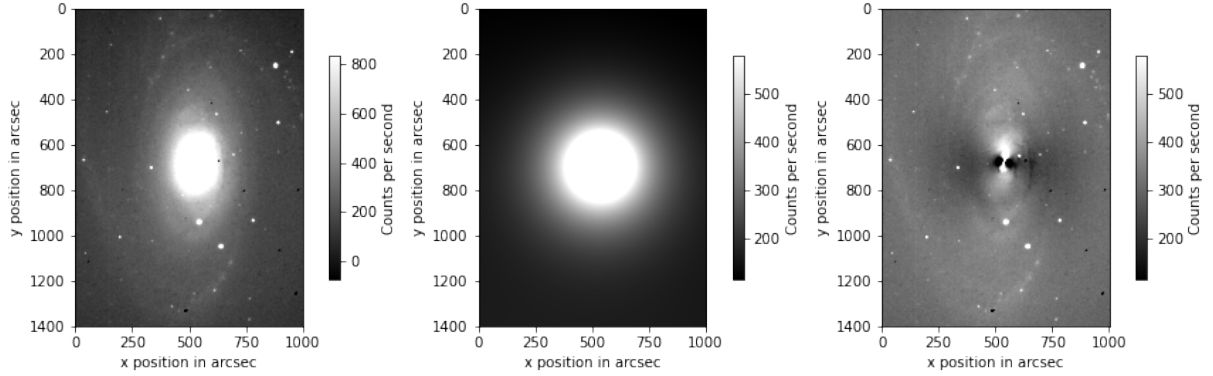


Figure 3: The Galfit result of the fitted Bode Galaxy. Here, one can see the galaxy, the fitted model and the residual.

The model for the Bode galaxy is the combination of two Sersic models.

Table 2: Sersic Model Parameters for the Bode Galaxy

Model	Integrated Magnitude	Effective Radius (pixels)	Sersic Index
Exponential Profile	4.4749 ± 0.0062	-	-
2nd Function	6.0403 ± 0.0078	31.0093 ± 0.2234	2.1119 ± 0.0073

3.3 M49

With the reference star and equation 2 we were then able to calculate the magnitude photometric zeropoint which we determined to be $m_{zp} = 16.8 \pm 0.05$.

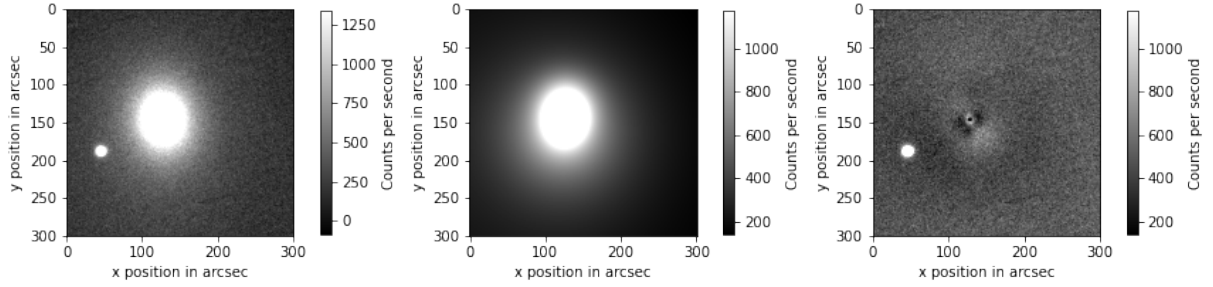


Figure 4: The Galfit result of the fitted M49 galaxy. Here, one can see the galaxy, the fitted model and the residual.

For this model we fitted three sersic models onto the galaxy. The results can be seen in table 3.

Table 3: Sersic Model Parameters for the M49 Galaxy

Model	Integrated Magnitude	Effective Radius (pixels)	Sersic Index
1st Function	7.1035 ± 0.0891	29.5026 ± 0.7432	0.5704 ± 0.0263
2nd Function	6.6420 ± 0.0417	12.2584 ± 0.2843	0.9338 ± 0.0146
3rd Function	4.7505 ± 0.1219	96.5838 ± 8.7986	1.1589 ± 0.0601

4 Discussion

4.1 Whirlpool Galaxy

As the first sersic function is the brightest and the effective radius is the smallest, we can assume, that it models the bulge. This is also reflected in the sersic indices of the functions. Here, the sersic index is 0.4117 ± 0.0107 , which coincides with the index of a

central bulge. As the sersic index of the second and the fourth function are 4.5496 ± 0.4748 and 3.9602 ± 0.2584 with rather large effective radii of 171.9334 ± 85.8887 pixels and 120.7271 ± 34.7358 pixels we can deduce from the data, that these two functions model the spiral arms of the galaxy. As the third sersic profile has the largest effective radius with 255.5811 ± 1.5991 pixels and a low sersic index of 0.1574 ± 0.003 we can also assume that it fits the fainter elliptical profile of the galaxy with its spiral arms. As we can clearly distinguish between the spiral arms of the galaxy and the bulge we are able to conclude that the galaxy has indeed a spiral form.

4.2 Bode Galaxy

Here it is not as clear to determine which morphologie the galaxy has. This can be due to the fact that the fit does not accurately resemble the galaxy. An indicator is the residual where the spiral arms are still very visible. As the bulge of the galaxy is way brighter than the spiral arms, which can also be seen in figure 3, the fit concentrated on that region. Figure 3 depicts a spiral galaxy with a brighter bulge. As the integrated magnitude of the second sersic function is higher than the first one, we can assume, that it fits the bulge, while the outer part is fitted by the first function. This is also reflected in the integrated magnitude of the second sersic fit is higher than that of the first one.

4.3 M49

Here, figure 4 depicts an elliptical galaxy. As the first two sersic functions have a higher magnitude and a lower effective radius we can assume, that they fit the center of the galaxy, while the third function fits the outer region. Here, even though visually the model seems to resemble the measured galaxy quite well, which we can also conclude from the residual, the sersic indexes disprove that. As they are all around 1 they suggest, that the galaxy is spiral shaped or a dwarf elliptical galaxy. This is not the case, as M49 is a elliptical galaxy [4]. A reason to why this fit is problematic could be the companion of the galaxy, which might have influenced the fitting of the functions.

5 Conclusion

We investigated the morphology of the galaxies M81, M51, and M49, using the HPP telescope and then fitted a Sersic profile using GalFit. The resulting models provided morphological parameters such as integrated magnitude, effective radius, and Sersic index, allowing for the classification of the galaxies into different morphological types.

The parameters show that M51 has characteristics of a spiral galaxies, with a bulge and spiral arms. The fitted models represent these features. M51 however seems not to be fitted quite as well, as the spiral arms are not included in the fit. M49 displayed the characteristics of an elliptical galaxy, with a central concentrated light distribution and a decline in brightness with increasing radius.

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